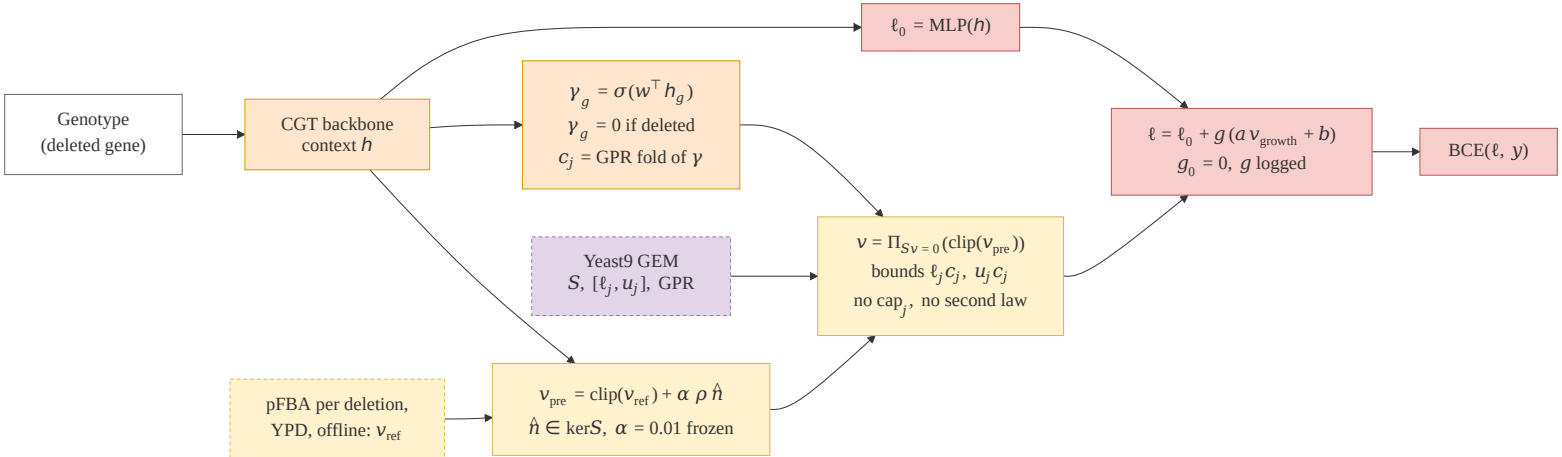

Genotype
(deleted gene)
CGT backbone
context h
$\ell_0 = \mathrm{MLP}(h)$
$\gamma_g = \sigma(w^\top h_g)$
$\gamma_g = 0$ if deleted
c_j = GPR fold of γ
Yeast9 GEM
S, $[\ell_j, u_j]$, GPR
pFBA per deletion,
YPD, offline: v_{ref}
$v_{\mathrm{pre}} = \mathrm{clip}(v_{\mathrm{ref}}) + \alpha\ \rho\ \hat{n}$
$\hat{n} \in \ker S$, $\alpha = 0.01$ frozen
$v = \Pi_{Sv=0}(\mathrm{clip}(v_{\mathrm{pre}}))$
bounds $\ell_j c_j$, $u_j c_j$
no cap$_j$, no second law
$\ell = \ell_0 + g\,(a\,v_{\mathrm{growth}} + b)$
$g_0 = 0$, g logged
BCE(ℓ, y)